

UNIVERSITY OF THE PHILIPPINES
Tacloban College
Division of Natural Sciences and Mathematics
Bachelor of Science in Computer Science

CMSC 12 — Fundamentals of Programming 2
Second Semester AY 2025-2026

MACHINE PROBLEM

Memoriam: Aspect of Fate **Technical Report**

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A. Rationale and Background

Memoriam is a top-down action roguelite game built entirely in Java. The player assumes the role of The Fool, a nameless wanderer tasked with restoring a world split in two. Each playthrough challenges the player to fight through four increasingly difficult levels, collect Arcana cards that permanently upgrade their abilities, choose a powerful Relic at the start of the run, and ultimately defeat the Final Boss.

The project was designed to demonstrate core Object-Oriented Programming principles in the context of an interactive real-time application. Rather than building a simple clone of a well-known game, the team designed Memoriam from the ground up, including all art assets, a custom game loop, a hand-rolled collision system, a card-based upgrade system, a save/load system, and a world renderer with camera scrolling. This scope allowed every major OOP concept, encapsulation, inheritance, polymorphism, and abstraction—to be applied in a meaningful, non-trivial way.

B. Objectives

This project aimed to:

- Apply core OOP principles—encapsulation, inheritance, polymorphism, and abstraction, across all major game systems.
- Build a scene management system that supports multiple distinct game states (menus, cutscenes, playable levels, pause screen) without restarting the application.
- Implement a custom rectangle-based collision detection system without relying on external physics libraries
- Create a card-driven upgrade system that persists player abilities and stats across level transitions within a single run
- Develop a file-handling system that saves and loads player progress between sessions. The system records cumulative statistics, including total completed runs and total enemies defeated, allowing player progression to persist across multiple playthroughs.
- Design a world renderer with camera logic that keeps the player centred while scrolling the map and all objects relative to the player's movement.
- Integrate an image library, GIF-based sprite animation, and an audio manager to produce a polished, responsive game feel.

C. Tools and Technologies

- | | |
|-------------|-------------|
| ● Procreate | - 2D Assets |
| ● Krita | - 2D Assets |

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- Dungeondraft - 2D Assets
- Visual Studio Code (1.114) - IDE
- Github (Bash and Desktop) - Version control and Online Collaboration
- Java (25.0.2) - Language to be used

D. Functional Requirements (Gameplay Dynamics)

Movement

The player moves using the WASD keys. Movement is processed every frame by the InputManager, which converts held keys into a direction vector. The vector is normalised and multiplied by the player's current speed stat to produce a per-frame displacement. Diagonal movement is clamped to the same maximum speed as axis-aligned movement, preventing a speed advantage.

The world scrolls around the player: once the player moves past a threshold from the screen centre, all game objects (map, enemies, projectiles, chests) translate in the opposite direction, giving the illusion of a scrolling camera while the player stays near the centre of the viewport.

Combat

The player shoots projectiles by clicking the mouse. Projectiles travel toward the mouse cursor position at a fixed speed. The fire rate is governed by a cooldown counter decremented each frame. Key combat features include:

- Single-shot by default; Multi Shot (Ten of Swords) increases projectile count to 3–5 in a spread pattern.
- Flame Shot (Ace of Wands) applies a percentage chance to deal bonus fire damage on each projectile.
- A universal crit system rolls against the player's crit chance stat for every projectile fired; a successful crit multiplies damage by 2×.
- Enemies also shoot projectiles (Purple/Yellow Wisps, Final Boss). Enemy projectiles damage the player on contact.
- Blue Wisps are melee enemies that deal damage on direct collision with the player rather than using projectiles.

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Progression & Level Structure

A standard run follows a fixed six-stage sequence:

Level ID	Name	Primary Content
0	Lobby (GameStart)	No enemies. Player selects a starting Relic from three options.
1	Level 1	Blue Wisps + Blue Mini-Boss.
2	Level 2	Purple Wisps + Purple Mini-Boss.
3	Level 3	Yellow Wisps + Yellow Mini-Boss.
4	Level 4	Mixed enemies + randomly selected Mini-Boss.
5	Boss Level	Final Boss + mixed wave spawns. Defeating the boss triggers the ending cutscene.

After all enemies in a level are defeated, the Treasure Chest unlocks. The player interacts with it (press E) to choose one of two randomly drawn Arcana cards. Selecting a card applies its ability and advances the run to the next level via a brief cutscene transition.

Win / Loss Conditions

- WIN: Defeat the Final Boss in Level 5. The ending cutscene plays and progress is recorded.
- LOSS: The player's HP reaches zero. If the Death Relic was chosen and has not been used, the player is resurrected once at 50% HP. Otherwise, the Lose screen displays and the save file is reset.

Card-Based Upgrade System

The game uses a card-based progression system built around Arcana cards. After clearing a level and opening the Treasure Chest, the player is presented with two randomly selected cards. Each card grants a specific upgrade that immediately affects gameplay. The available cards provide different effects:

- HP Regen restores health automatically over time.
- Fortified Regen increases survivability by combining regeneration and defensive bonuses.
- Multi Shot (Ten of Swords) increases the number of projectiles fired at once.
- Flame Shot (Ace of Wands) gives projectiles a chance to become fire projectiles.
- Shield grants temporary protection from damage.
- Speed Enhance increases movement speed.

The system also supports stacking upgrades. Selecting the same card multiple times strengthens its effect up to a maximum limit. Because cards are randomly generated every run, players are encouraged to adapt their builds depending on the upgrades they receive.

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Relic System

At the start of every run, the player selects one Relic from the “Big Three” inside the Lobby. Unlike Arcana cards, Relics act as major run-defining modifiers that stay active for the entire playthrough. The available Relics are:

- Death – Revives the player once upon death with half HP remaining.
- The Empress – Temporarily freezes enemies while disabling player movement during activation.
- The Magician – Improves offensive capabilities through stronger projectile behaviour.

Relics are designed to support different playstyles. Death focuses on survivability, The Empress provides crowd control, while The Magician strengthens combat performance.

Save and Load System

The game includes a persistent save system that stores player progress between sessions. Progress is recorded whenever the player clears levels or defeats the Final Boss. The save system stores:

- Current level progress
- Player HP
- Active abilities and upgrades
- Total enemies defeated
- Completed runs

When the application is launched again, the player may continue from the latest saved state through the Load option in the menu. If the player loses without a remaining revive effect, the current run is reset while cumulative statistics remain saved separately. This system allows long-term progression data to persist across multiple playthroughs while still maintaining the roguelite structure of restarting runs after defeat.

Infinite Mode

The game includes an Infinite Mode that functions separately from the main six-level progression system. Instead of progressing toward a final boss, the player survives against endless enemy waves for as long as possible. In this mode:

- Enemies continuously spawn throughout the game.
- The kill counter tracks the player’s total eliminations.
- Difficulty increases as more enemies are defeated.
- Gameplay only ends when the player dies.

Infinite Mode was designed to improve replayability while also testing the stability of the combat, collision, rendering, and enemy spawning systems during extended gameplay sessions.

E. Methodology

Asset Production

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All primary visual effects—player sprites (idle, attack, shield), enemy sprites (Blue, Yellow, Purple, and Bosses), UI buttons, Cards, Background Maps, and HUD elements—were produced by the development team. Animated sprites are in GIF files. Audio assets are all sourced from Pixabay and Felisyan Studios except for the Main Menu background music which was also produced by the development team.

Game Loop

The game uses a custom real-time game loop built entirely in Java Swing. The loop continuously processes input, gameplay logic, collision detection, and rendering every frame. During each update cycle:

- Keyboard and mouse inputs are processed.
- Player, enemies, and projectiles update their positions and behaviour.
- Collision checks are performed between objects.
- Enemy AI updates movement and attack patterns.
- The renderer redraws the map and all active objects.

The game also uses interpolation for smoother rendering. Instead of instantly snapping objects to their updated positions, rendered positions gradually move toward the target coordinates, creating smoother movement on screen.

Class Hierarchy and OOP Design

The project heavily applies Object-Oriented Programming principles across all gameplay systems. Shared behaviour is grouped into reusable structures while specialised objects extend their own unique functionality. All interactive objects share common features such as:

- Position handling
- Movement
- Rendering
- Scaling
- Collision support

From these shared systems, specialised behaviours are built for:

- Player movement and combat

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- Enemy AI and attacks
- Projectiles
- Treasure Chests
- Maps and barriers
- Boss encounters

Inheritance reduces code duplication by allowing shared systems to be reused across multiple object types. Polymorphism is applied through overridable update and collision behaviour, while encapsulation is maintained through separated gameplay systems such as audio, saves, status effects, and upgrades.

Scene and State Management

The game uses a scene-based structure that separates menus, gameplay, cutscenes, and overlays into independent states. This allows smooth transitions between different parts of the game without restarting the application. Implemented scenes include:

- Main Menu
- Lobby
- Gameplay Levels
- Pause Screen
- Lose Screen
- Credits Screen
- Loading Screens
- Cutscenes

This structure simplifies scene transitions and prevents gameplay systems from interfering with menu systems. Because scenes are modular, additional screens and levels can be added without major changes to the overall architecture.

Stat and Modifier Architecture

The game uses a modular stat system to manage player attributes and upgrade effects dynamically. Core gameplay values such as HP, attack damage, movement speed, projectile count, and

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regeneration are updated through modifiers rather than hardcoded values. Whenever the player gains a card ability, the corresponding modifiers are applied automatically. This allows upgrades to affect gameplay immediately without rewriting combat logic. Examples include:

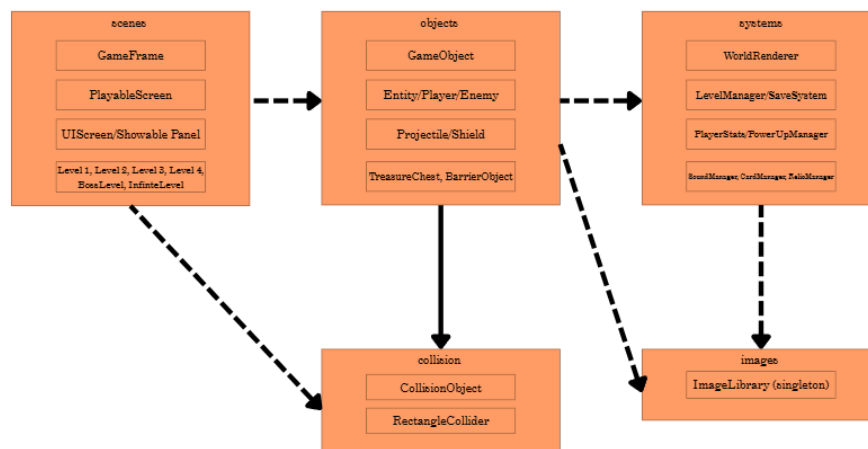
- Multi Shot increasing projectile count.
- Speed Enhance increasing movement speed.
- HP Regen lowering regeneration intervals.
- Flame Shot modifying projectile behaviour.

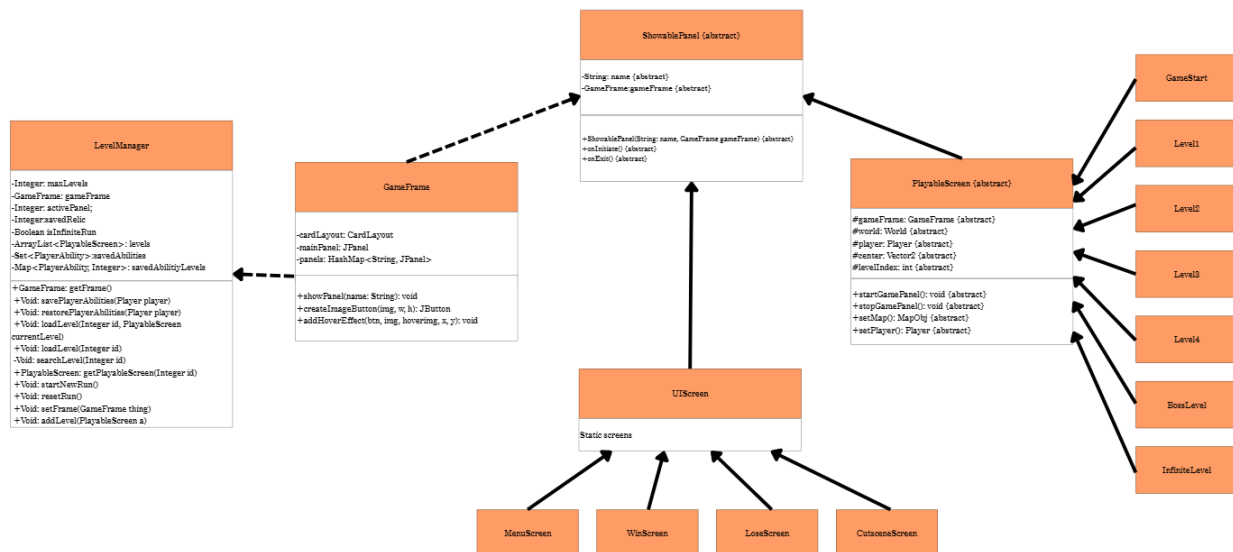
Temporary effects such as shields and regeneration boosts are handled separately from permanent stat modifiers. This separation keeps the system flexible and easier to expand for future upgrades and abilities.

UML Diagram

Figure 1.1 (High-level Architecture Diagram)

High-level Architecture





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Figure 4. (Systems Package Diagram)

Systems



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F. Results

The following describes the completed game state as verified during final testing.

Main Menu

An animated main menu displays “The Fool” on a cliff in a custom background, inspired by the actual “Fool” card, with five image based buttons: Start, Practice, Tutorial, Credits, and Exit. Background music begins immediately on menu load.

Figure 1. Main Menu



Lobby & Relic Selection

The Lobby level loads with ambient music and the player spawns in the map. The treasure chest is immediately visible and interactable. Pressing E opens the Big Three Relic card overlay showing Death, The Magician, and the Empress with full artwork and hover highlights.

Figure 2.1. Lobby

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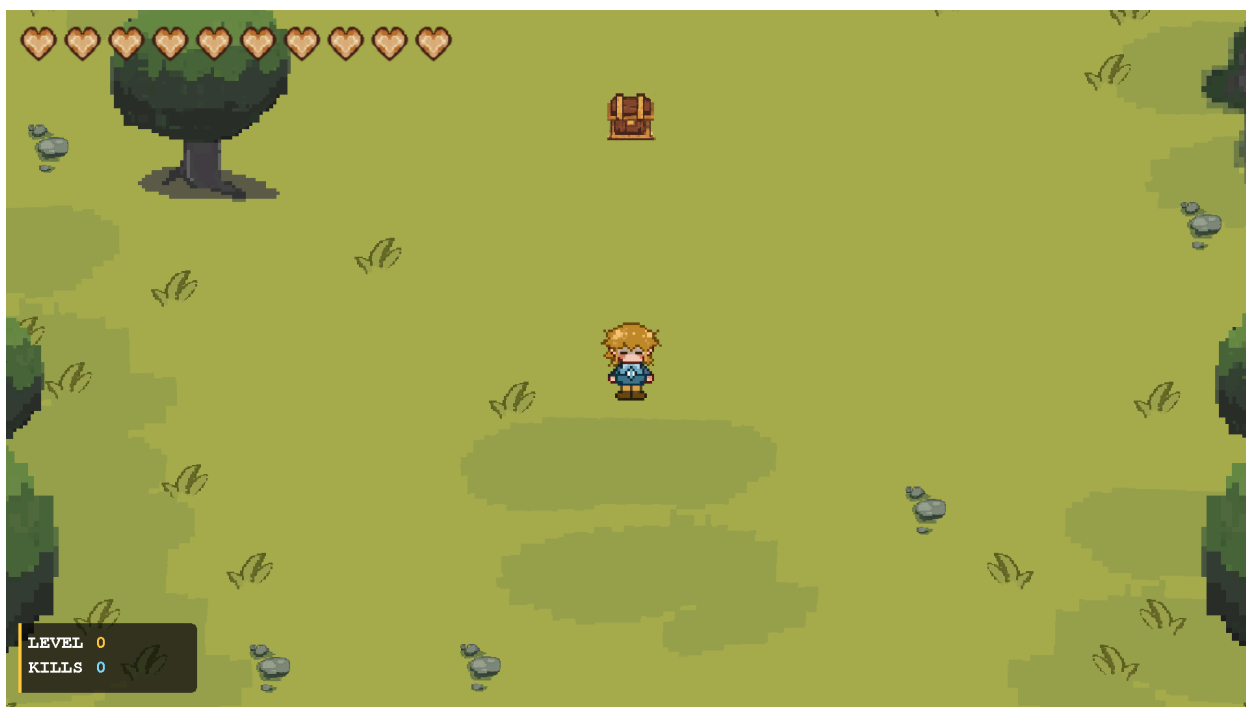
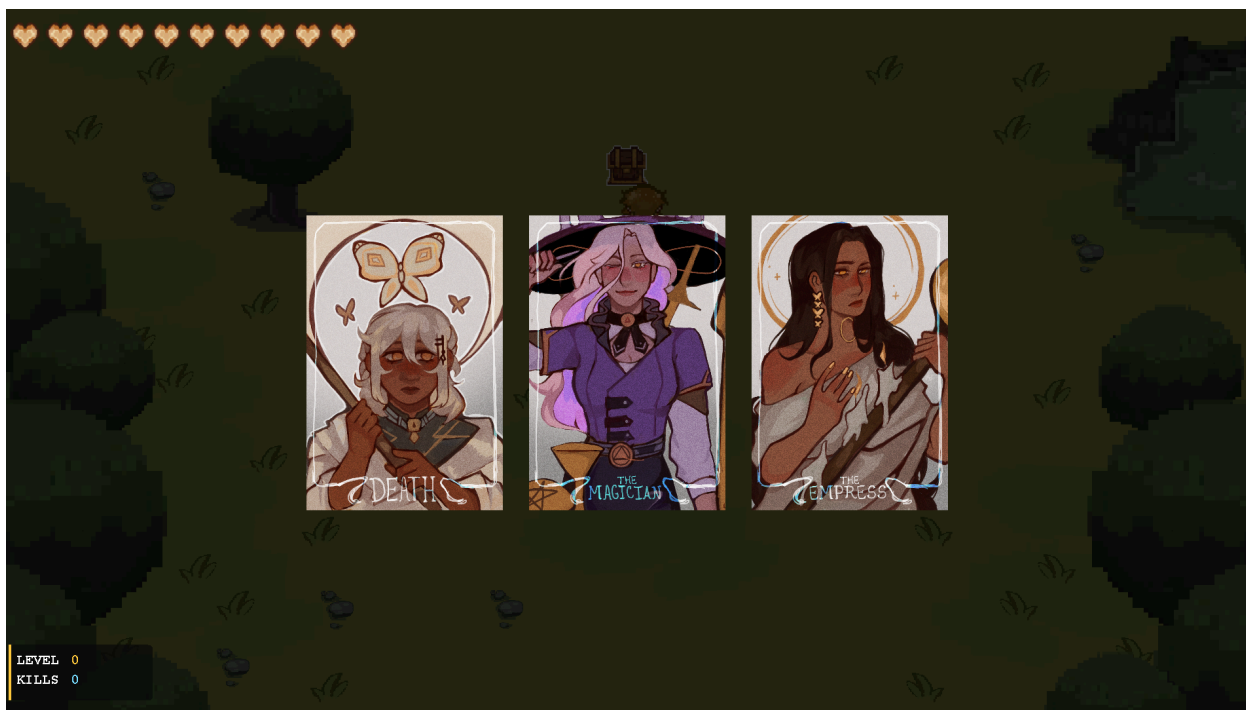


Figure 2.2. Big Three Relic



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Gameplay: Level 1

The background music starts immediately, and the player spawns into the map with all hearts filled and the relic that was chosen by the player. Each collision with an enemy or an enemy projectile results in a loss of one heart. Alternatively, the player can shoot projectiles that can kill an enemy upon collision.

Figure 3. Gameplay: Level 1



Card Selection

After clearing a level and interacting with the Treasure Chest, the screen darkens and presents two randomly generated cards based on their rarity. Cards animate briefly (slot machine spin) before settling. Hovering a card displays its name and its skills: clicking confirms its selection and triggers the next level.

Figure 4. Card Selection

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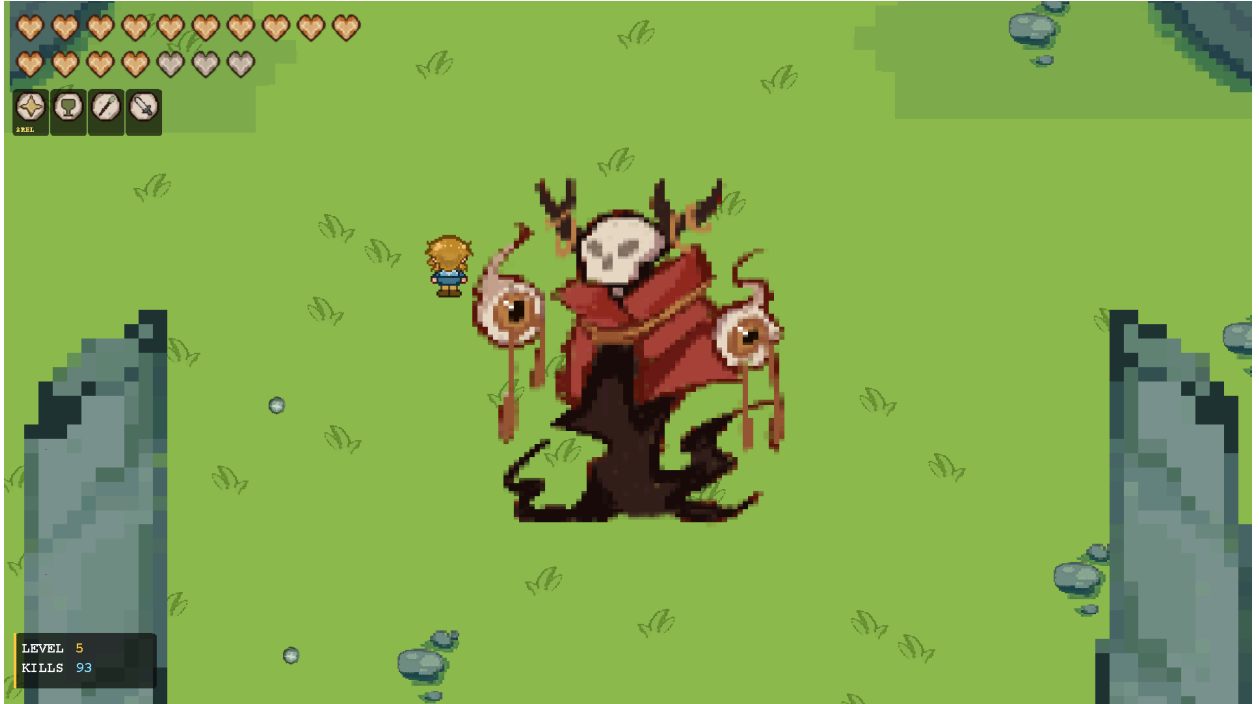


Boss Level

The boss has two phases. In Phase 1, the boss does not move, and can shoot fireballs while spawning blue minions. In Phase 2, the boss can move around the player, shooting fireballs at an increased rate while spawning blue and purple minions.

Figure 5. Boss Level

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HUD and Status Icons

The HUD consistently renders the Heart Bar in the upper-left corner of the screen. Below it, ability icons appear if it has been selected. Relic icons display with a REL label. The kill counter and the level number are shown in the lower-left corner of the screen.

Figure 6. HUD and Status Icons

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G. Problems/Bugs

The following known issues were identified during development and testing.

UI/UX

- **Main Menu Button Scaling:** On devices using display scaling settings such as 125% or 150%, the main menu buttons may appear misaligned or incorrectly scaled. This occurs because the `JPanel` still reports a resolution of 1920×1080 even when the actual displayed size is scaled larger by the operating system, affecting the positioning and scaling calculations of the UI elements.

Autosaving System

- The Autosave System does not work unless the application is run in administrator mode.

Assets Folder Dependency

- The application requires the jar (or exe file) to be in the same folder as APPDATA

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and Assets folder in order to load applications.

Command Line Run

- The bin files must be run in "The F4tes/Memorial" directory

Shooting Bug/Feature

- Once the player shoots, it nudges the player object by a few pixels in the current movement direction; this is initially a bug turned into a fun speedrunning mechanic.